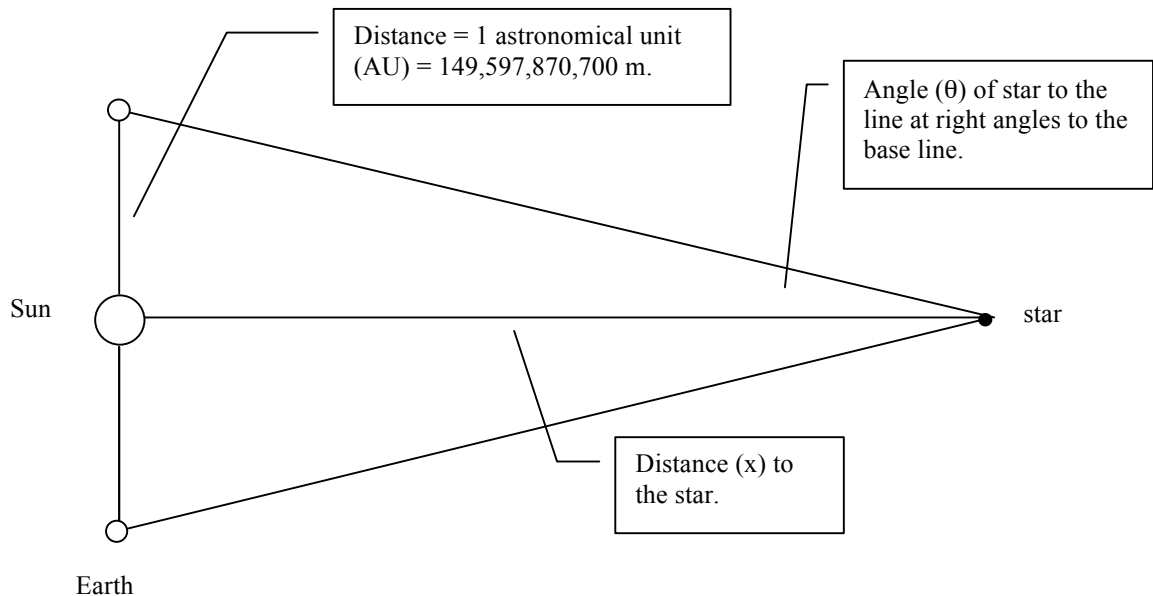


MEASURING DISTANCES TO STARS

Close Stars

- **Parallax method** is used.
- Stars are observed 6 months apart.
- Close stars appear to move but distant stars do not.
- Trigonometry is used to calculate the distance.



To calculate the distance x to a star :

$$\tan \theta = \frac{149,597,870,700 \text{ m}}{x}$$

Assume the parallax angle measured to a star is 0.0136° .

$$\Rightarrow \tan 0.0136^\circ = \frac{149,597,870,700}{x}$$

$$\Rightarrow \underline{x = 6.032 \times 10^{14} \text{ m}}$$

- The **parsec** (pc) is used as a measure of intergalactic distances.
- Using the above calculation, with parallax angle $q = 1.0$ second of arc ($1/3600^\circ$):

$$\mathbf{1.00 \text{ parsec} = 3.085678 \times 10^{16} \text{ m} = 3.261564 \text{ light years}}$$

Distant Stars

- Distant stars do not appear to move very much when viewed by parallax method.
- The ***apparent brightness*** of the stars is used to measure their distances.
- ***Cepheid variables*** vary in brightness over a set period of time and are used to estimate distances to galaxies.